

RESEARCH LETTER

Accuracy of a Smartwatch-Derived ECG for Diagnosing Bradyarrhythmias, Tachyarrhythmias, and Cardiac Ischemia

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The Apple Watch Series 4 and 5 (Apple Inc, Cupertino, CA) allows wearers to record single-lead electrocardiograms (AW-ECGs). This ability to self-register ECGs and to digitally send them to healthcare providers could prove valuable in settings of resource constraints, limited access to health care, and during periods of physical distancing or quarantines. In fact, the Food and Drug Administration recently issued an enforcement policy expanding the capability of such noninvasive remote monitoring devices to facilitate patient monitoring during the coronavirus disease 2019 pandemic.¹ While the smartwatch will only indicate whether there are signs of atrial fibrillation (AF) present, the technology could also prove useful for the detection of other clinically important cardiac abnormalities.^{2–4} We, therefore, sought to evaluate the accuracy of AW-ECGs interpreted by cardiologists for diagnosing bradyarrhythmias, tachyarrhythmias, and cardiac ischemia.

To capture a wide variety of ECG findings, 256 patients with or without a history of cardiovascular disease were enrolled from outpatient or emergency departments. The study was approved by our institutional review committee and all subjects provided informed consent. Standard 12-lead ECGs were performed, followed by smartwatch ECG recordings with the watch worn on the left wrist. Patients received brief instructions on how to register a smartwatch ECG and were then asked to independently record a single 30-second tracing. Two blinded cardiologists independently evaluated all 12-lead ECGs and 2 different blinded cardiologists evaluated all AW-ECGs. Diagnostic disagreements were adjudicated by a third

blinded cardiologist. ECGs were classified into one or more of the following groups:

1. Bradyarrhythmia (<55 bpm): second- or third-degree atrioventricular block (AVB, n=13), sinus bradycardia (n=27).
2. Tachyarrhythmia (≥100 bpm): AF (n=49), atrial flutter/tachycardia (n=13), ventricular tachycardia (n=2).
3. Ischemia-related abnormality: pathological Q-waves, abnormal ST-segment, or T waves (n=81).
4. No abnormalities: normal sinus rhythm without bradyarrhythmia, tachyarrhythmia, or ischemia-related abnormality (n=103).

Cohen's kappa (κ) statistics were calculated for the classification of AW-ECGs as abnormal (groups 1–3) versus no abnormalities. Using the 12-lead ECG as the gold standard, the diagnostic accuracy of cardiologist-interpreted AW-ECGs was expressed as sensitivity and specificity.

The patients' mean age was 66±6 years; 58% were male. As patients self-registered AW-ECGs, baseline wander and artifacts were commonly introduced, but all tracings were judged to be interpretable. Cardiologists 1 and 2 demonstrated high levels of agreement when classifying AW tracings as abnormal ($\kappa=0.85$ [95% CI, 0.74–0.96]). Using AW-ECG, 35 abnormalities were incorrectly labeled as no abnormalities (14%), most of which were due to pathological Q-waves (n=17) or ST/T-wave abnormalities (n=13) seen in other leads on the 12-lead ECG. Using AW-ECG, all 40 patients with bradyarrhythmia were identified. Atrioventricular block

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Nonstandard Abbreviations and Acronyms

AF	atrial fibrillation
AW-ECGs	Apple Watch ECGs

and sinus bradycardia were correctly differentiated in 81% of patients. AW-ECG identified all 64 patients with tachyarrhythmia. The 2 patients with VT were correctly detected using AW-ECG, whereas AF and atrial flutter/tachycardia were correctly differentiated in 71% of cases. Diagnostic accuracy was high for AF (sensitivity, 96% [95% CI, 86%–99%]; specificity, 91% [95% CI, 87%–95%] with 9/19 false positives being atrial flutter/tachycardia) but low for atrial flutter/tachycardia (sensitivity, 25% [95% CI, 5%–57%] and specificity, 99% [95% CI, 97%–100%]). The ability of AW-ECG to detect abnormalities related to cardiac ischemia was limited (sensitivity, 7% [95% CI, 1%–24%] and specificity, 100% [95% CI, 98%–100%] for Q-waves; sensitivity, 34% [95% CI, 21%–49%] and specificity, 100% [95% CI, 97%–100%] for ST/T-wave abnormalities). A common cause of misdiagnoses was due to the AW-ECG being normal in the setting of a previous anterior transmural infarct (Figure).

Our results demonstrate that Apple Watch ECG tracings can provide clinically relevant information beyond detecting AF. Bradyarrhythmias and tachyarrhythmias were accurately detected and the specific rhythm correctly diagnosed in most patients. Atrial rhythms were less accurately diagnosed because the smartwatch records the equivalent of lead I, which can be nearly perpendicular to the normal atrial depolarization vector. This technology has high specificity but poor sensitivity for ischemic heart disease. Accordingly, a normal AW-ECG tracing should not be used to reassure patients or health care professionals when it is suspected. The major limitation of Apple Watch tracings is that single-lead ECGs provide a limited view of cardiac electrical activity, thus impacting their ability to detect regional abnormalities. Diagnostic accuracy may be improved by recording multiple AW-ECGs from unconventional positions, including the left ankle or chest wall.²

In the first quarter of 2020, Apple Inc shipped millions of smartwatches worldwide, representing a 23% increase in sales from the previous year. As an increasingly popular direct-to-consumer device, physicians will

increasingly be asked to interpret acquired data. Our study suggests that this technology has the potential to empower patients to play more active roles in their cardiac care and that it can be an important tool in clinical practice. Future studies are needed to evaluate patients' ability to register smartwatch ECGs at home over long periods and the added benefit of recording smartwatch single-lead tracings from unconventional locations. These data could have important implications for the feasibility and full potential of cardiac monitoring strategies that incorporate this technology, thus informing its integration into the evolving practice of telemedicine.

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Disclosures

None.

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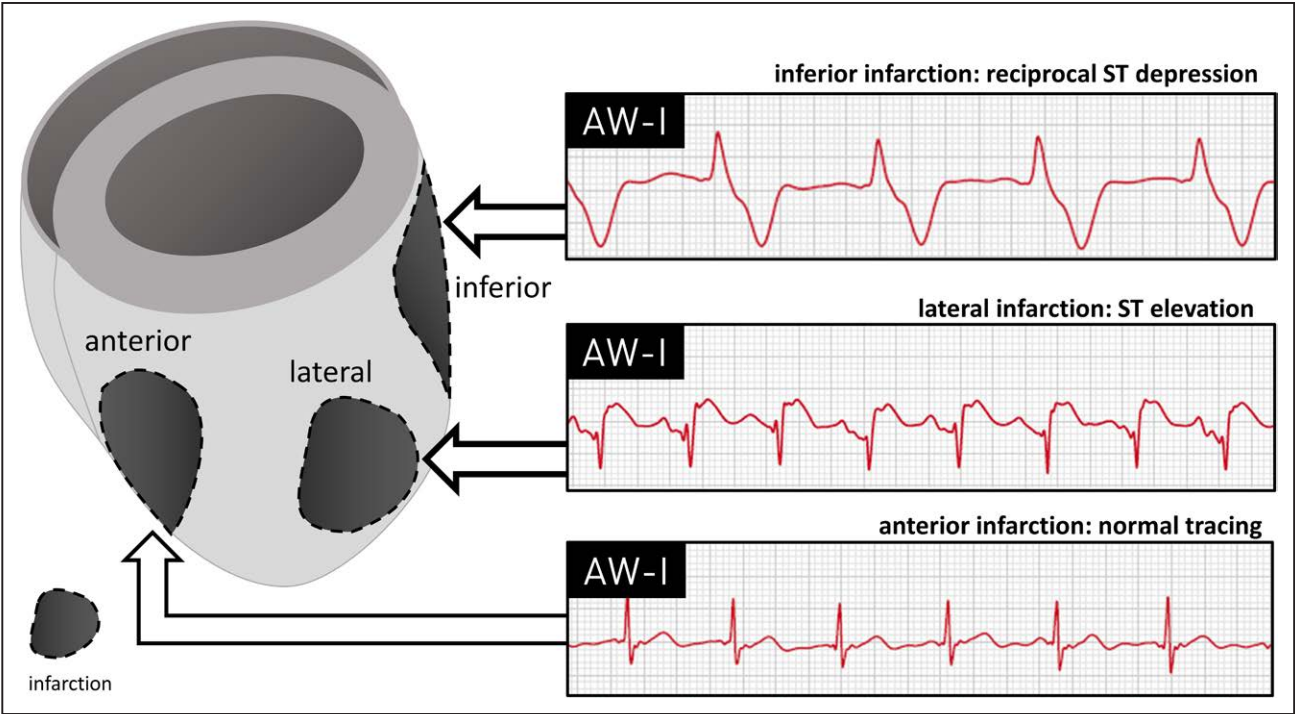


Figure. Typical examples of ST-segment–elevation myocardial infarctions in different coronary distributions with corresponding Apple Watch tracings. While ST depression suggests reciprocal changes in inferior infarction and ST-segment elevation suggests lateral infarction, the tracing may also be entirely normal in case of anterior infarction.